

EDUCATION

Sapienza University of Rome
Sep 2023 – Oct 2025

MS Economics and Finance – 110 cum laude

- Focus on quantitative methods and financial econometrics
- Thesis: Asset Pricing with Machine Learning
Supervisor: Prof. Giuseppe Ragusa

Sapienza University of Rome
Sep 2020 – Oct 2023

BS Economics

- Focus on Economic Theory
- Thesis: Dedollarization of the International Monetary System
Supervisor: Prof. Marco Di Pietro

PROJECTS

Asset Pricing with Machine Learning

Applying machine learning to stock return forecasting

- Replicated and extended Gu, Kelly & Xiu (2020) using a modern sample
- Worked with a large dataset of over 3 million firm-month observations
- Compared neural networks with linear models
- Implemented the full modelling workflow in Python, from data processing to model validation

Forecasting Macro Trends with Factor Models

Estimation of macroeconomic factors via PCA with missing data

- Used the FRED-MD dataset (McCracken & Ng)
- Identified 7 latent factors explaining 47.36% of total variance up to March 2025
- Produced one-step-ahead forecasts of inflation and industrial production

Volatility Forecasting

Comparison of models for Euro Stoxx 50 volatility

- GARCH model chosen as benchmark for volatility prediction
- Multilayer Perceptron outperformed all models in out-of-sample forecasting

Identifying Macroeconomic Shocks

Quantitative assessment of macroeconomic shocks using VAR and LP

- Explored multiple identification strategies based on influential macroeconomic papers
- Estimated impulse responses and compared results across VAR and Local Projections

Portfolio Construction

Markowitz Model to find the Optimal Portfolio

- Constructed sector-level portfolios and optimised each independently to identify top-performing assets
- Combined assets from sector portfolios into a single global portfolio and re-optimised it
- Observed an increasing weight on the Health sector in recent years

SKILLS & INTERESTS

Technical Skills

- Python (NumPy, pandas, scikit-learn, statsmodels, matplotlib) for data analysis, modelling and research
- R for statistical analysis and econometrics
- Git and GitHub for version control and project documentation

Leadership & Teamwork

- Curious and resilient, with initiative and a strong sense of responsibility. Strong communicator and listener. Leads when needed, always a team player

Languages

- Italian (native); English (fluent); Spanish (intermediate); French (intermediate)